# Chimera States in Coupled Phase Oscillators

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**Abstract:** Chimera states are unusual spatiotemporal patterns in which an array of identical oscillators spontaneously splits into two coexisting domains: one coherent and phase-locked, the other incoherent and desynchronized. First found numerically by Kuramoto and Battogtokh [1] and analyzed exactly by Abrams and Strogatz [2], chimera states arise only under nonlocal coupling and cannot occur with purely local or purely global topologies. We simulate the Abrams–Strogatz model on a ring of $N = 512$ phase oscillators coupled through a cosine kernel $G(x) = (1 + A\cos x)/(2\pi)$ with $A = 0.995$ and phase-lag $\alpha = \pi/2 - 0.18$. We reproduce the key diagnostics from the original paper (the local coherence $R(x)$, local average phase $\Theta(x)$, and mean drift frequency $\Delta(x)$) and perform a bifurcation sweep over $\alpha$ to find the critical threshold at $\alpha \approx 1.38$ where the system transitions from full synchronization ($\langle R\rangle \approx 1$) to a stable chimera ($\langle R\rangle \approx 0.6$). An automated chimera-survival mechanism ensures reliable nucleation from random initial conditions.

***Keywords:*** chimera states, coupled oscillators, nonlocal coupling, phase dynamics, spontaneous symmetry breaking, Kuramoto model

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## 1. Introduction

The study of coupled oscillators has a long history in nonlinear dynamics, starting with Huygens' observation of sympathetic resonance between pendulum clocks in 1665 and later formalized by Winfree [3] and Kuramoto [4]. A fundamental question in this field is what collective behaviors can emerge from a population of interacting oscillators.

For *non-identical* oscillators with distributed natural frequencies, partial synchronization is well understood. Fast oscillators lock together while slow ones drift, and the boundary between the two populations is set by the ratio of frequency spread to coupling strength [4], [5]. The split is a direct result of heterogeneity among the oscillators.

However, in 2002, Kuramoto and Battogtokh [1] discovered that a ring of *identical* oscillators could spontaneously break its own symmetry and split into coexisting coherent and incoherent domains. The phenomenon (later named a "chimera state" by Abrams and Strogatz [2]) cannot be attributed to any difference between the oscillators. Every oscillator follows the same equation with the same parameters, yet some lock together while others drift, and the pattern persists indefinitely as a stable state.

Abrams and Strogatz [2] gave the first exact solution for chimera states, analyzing a ring of phase oscillators coupled through a cosine kernel. They showed that the stable chimera bifurcates from a spatially modulated drift state and is destroyed in a saddle-node bifurcation with an unstable chimera, tracing out the precise region of parameter space where chimeras can exist.

The key ingredient is **nonlocal coupling**, a topology intermediate between nearest-neighbor (local) and all-to-all (global). Local coupling constrains spatial patterns through diffusive smoothing. Global coupling gives every oscillator the same mean field, leaving no spatial structure to break. Nonlocal coupling provides

[1] Source code and report: **https://github.com/Darsh-A/Chimera_NLD**

just enough spatial variation to sustain coexisting coherent and incoherent domains [2], [6].

Since then, chimera states have been observed in chemical oscillators [7], mechanical metronomes, and optical systems. They have also been connected to unihemispheric sleep in certain animals, where one brain hemisphere stays awake while the other sleeps [8].

**Scope of this project.** We simulate the Abrams–Strogatz model on a ring of $N = 512$ coupled phase oscillators, reproduce the key diagnostic plots from the original paper [2], and sweep the phase-lag parameter $\alpha$ to map the transition from full synchronization to the chimera regime. All code and the compiled report are available at **https://github.com/Darsh-A/Chimera_NLD**.

## 2. Mathematical Model

### 2.1 Governing Equation

The simplest system known to support chimera states is a ring of phase oscillators governed by the integro-differential equation [1], [2]:

$$\frac{\partial \varphi}{\partial t} = \omega - \int_{-\pi}^{\pi} G(x - x') \times \sin[\varphi(x,t) - \varphi(x',t) + \alpha]\, dx' \tag{1}$$

Here $\varphi(x,t)$ is the phase of the oscillator at position $x$ on the ring at time $t$, with $x \in [-\pi, \pi]$ and periodic boundary conditions. The natural frequency $\omega$ is the same for all oscillators; one can set $\varphi \to \varphi + \omega t$ to eliminate it entirely, so it plays no dynamical role. The parameter $\alpha \in [0, \pi/2]$ is a tunable phase lag in the interaction function.

### 2.2 Coupling Kernel

The kernel $G(x - x')$ controls how strongly oscillators at different positions interact. Following Abrams and Strogatz [2], we use the cosine kernel:

$$G(x) = \frac{1 + A\cos x}{2\pi} \tag{2}$$

where $0 \le A \le 1$. This kernel is even, non-negative, peaks at $x = 0$ (strongest coupling between nearby oscillators), and integrates to one over the ring. When $A = 0$ the coupling is all-to-all; when $A = 1$ neighboring oscillators are coupled twice as strongly as diametrically opposite ones. The cosine form was chosen because it admits an exact analytical solution of the self-consistency equations, unlike the exponential kernel originally used by Kuramoto and Battogtokh [1].

### 2.3 Complex Order Parameter

To measure local synchronization, Abrams and Strogatz introduce a space-dependent complex order parameter [2]:

$$R(x,t)e^{i\Theta(x,t)} = \int_{-\pi}^{\pi} G(x - x')e^{i\theta(x',t)}\, dx' \tag{3}$$

where $\theta = \varphi - \Omega t$ is the phase measured in a frame rotating at frequency $\Omega$. The function $R(x)$ measures local coherence: $R(x) \approx 1$ means oscillators near position $x$ are well-synchronized, while $R(x) \ll 1$ indicates incoherence. The function $\Theta(x)$ gives the local mean phase. In the steady chimera, both $R$ and $\Theta$ depend on space but not on time.

For the cosine kernel, the exact solution takes the form [2]:

$$R(x)^2 = |c|^2 + 2\,\mathrm{Re}(ca^*)\cos x + |a|^2\cos^2 x \tag{4}$$

$$\tan\Theta(x) = \frac{c_i + a_i\cos x}{c_r + a_r\cos x} \tag{5}$$

where $c$ and $a$ are coefficients determined self-consistently. This is why $R(x)$ comes out with a cosine-like shape in the simulation.

### 2.4 Discrete Approximation

For numerics, we place $N$ oscillators at equally spaced positions $x_j = -\pi + 2\pi j/N$ for $j = 0, 1, \ldots, N-1$. The integral in Equation 1 becomes a Riemann sum with spacing $\Delta x = 2\pi/N$:

$$\frac{d\theta_i}{dt} = \omega - \frac{1}{N}\sum_{j=0}^{N-1}\left(1 + A\cos(x_i - x_j)\right) \times \sin\left(\theta_i - \theta_j + \alpha\right) \tag{6}$$

The $1/(2\pi)$ in the kernel normalization cancels with $\Delta x = 2\pi/N$, leaving $1/N$ as the overall prefactor. This is the system of $N$ coupled ODEs we integrate.

### *2.5 Parameters*

Following [2], we use $A = 0.995$ (strongly non-local coupling) and define $\beta = \pi/2 - \alpha$, so $\alpha = \pi/2 - \beta$. The paper's Figure 1 uses $\beta = 0.18$, giving $\alpha \approx 1.391$. We use $N = 512$ oscillators (the paper uses $N = 256$; the larger $N$ extends the chimera's metastable lifetime).

## 3. Numerical Methods

### *3.1 Initial Conditions*

The initial phases are drawn from a Gaussian-bump perturbation centered at $x = 0$:

$$\varphi_0(x) = a \cdot \exp(-\kappa x^2) \cdot r \tag{7}$$

where $r$ is uniform random on $[-1/2, 1/2]$, $a = 6$ is the amplitude, and $\kappa$ controls the width. The original paper uses $\kappa = 30$ (a narrow spike) with a fixed-step RK4 integrator at $\delta t = 0.025$. For our adaptive RK45 solver, we found that $\kappa = 0.76$ (a much wider bump) nucleates the chimera more reliably. The adaptive solver's variable step size tends to smooth out very localized features during the first few time steps, so the wider initial kick is needed to seed incoherence over a large enough spatial extent.

### *3.2 Time Integration*

We integrate Equation 6 using SciPy's `solve_ivp` with the Dormand–Prince RK45 method. The key settings are listed in Table 1.

| Parameter | Value |
|---|---|
| Method | RK45 (adaptive) |
| Time span | $t \in [0, 5000]$ |
| Output points | 1000 |
| Max step size | 0.1 |

*Table 1: Time-integration settings.*

The maximum step of 0.1 is much smaller than the default and close to the paper's fixed $\delta t = 0.025$. This ensures the fast phase dynamics during chimera formation are adequately resolved while still benefiting from adaptive error control.

### *3.3 Chimera Survival Check*

Chimera states in finite-$N$ systems are metastable: they can collapse to full synchronization after a random (but typically long) time. Not every initial condition successfully nucleates a chimera either. We handle both issues with an automated retry mechanism:

1. Generate a random initial condition from Equation 7.
2. Run a short **probe simulation** to $t = 500$ (10% of the full run).
3. Compute the global order parameter $R = |1/N \sum_j e^{i\theta_j}|$ at the probe endpoint.
4. If $R > 0.93$, the chimera has collapsed; discard and retry with a new random seed.
5. If $R < 0.93$, the chimera is alive; proceed with the full $t = 5000$ simulation.
6. Repeat up to 10 attempts.

This way the expensive full simulation only runs when the chimera has actually nucleated.

### *3.4 Diagnostics*

We compute the following quantities from the simulation output:

- **Sliding-window local order parameter** $|z(x)|$: For each oscillator $i$, we average $e^{i\theta_j}$ over a window of $2w + 1 = 81$ neighbors (periodic boundary) and take the modulus. Values near 1 indicate local coherence; values $\ll 1$ indicate incoherence.

- **Kernel-weighted order parameter** $R(x)$ and $\Theta(x)$: Computed directly from Equation 3 using the coupling kernel $G$. This matches the paper's definition exactly.

- **Mean drift frequency** $\langle \dot{\theta}_i \rangle$: Time-averaged angular velocity of each oscillator, computed from the unwrapped phase difference over the second half of the simulation. Locked oscillators share a common frequency; drifting ones have a spatially varying frequency.

- **Global order parameter** $\langle R \rangle$: Time-averaged value of $|1/N \sum_j e^{i\theta_j}|$ over the steady-state portion, used for the bifurcation sweep.

## 4. Results and Analysis

### *4.1 Phase Evolution*

Figure 1 shows snapshots of the oscillator phases at five key time-steps, capturing the

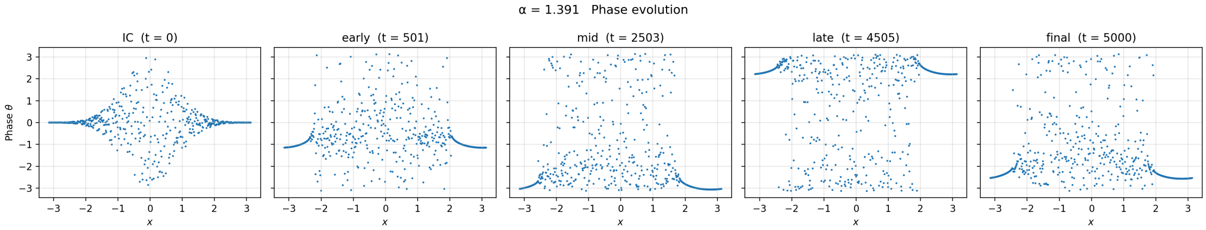


*Figure 1: Phase snapshots at five key times. From left to right: Gaussian-bump initial condition ($t = 0$), early chimera nucleation ($t = 501$), developing chimera ($t = 2503$), mature chimera ($t = 4505$), and final steady state ($t = 5000$). Parameters: $N = 512$, $A = 0.995$, $\alpha = 1.391$.*

evolution of the chimera from nucleation to its steady state.

At $t = 0$, the initial condition shows a spread of phases near $x = 0$, with all other oscillators starting near $\theta = 0$. By $t = 501$ the chimera has formed: oscillators near $x \approx \pm\pi$ have locked onto a common phase (the smooth arc), while those in the middle have started to drift apart. The middle panels ($t = 2503$, $t = 4505$) show the mature chimera, with a clearly defined coherent arc coexisting with a scattered cloud of drifting oscillators. The final panel confirms that by $t = 5000$ the state is statistically steady.

The locked oscillators slow down as they pass through the coherent pack, which is why the scattered dots cluster more densely near the coherent arc, a feature also noted in [2].

### 4.2 Local Order Parameter

Figure 2 shows the sliding-window local order parameter $|z(x)|$ at the final time-step, giving a direct, model-independent measure of local synchronization at each ring position.

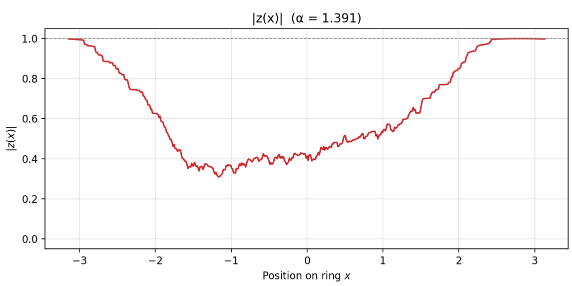


*Figure 2: Sliding-window local order parameter $|z(x)|$ at $t = 5000$. Window size: $2 \times 40 + 1 = 81$ oscillators.*

The plot shows the expected chimera structure:

- $|z(x)| \approx 1.0$ near $x \approx \pm\pi$: oscillators are fully phase-locked.
- $|z(x)|$ drops to roughly 0.35 in the incoherent region around $x \approx -1$, indicating a nearly uniform phase distribution.

The jaggedness in the incoherent region is a finite-$N$ effect: with only 81 oscillators in the averaging window, statistical fluctuations in the random phases produce small variations in $|z|$. These would shrink with larger $N$ or wider windows.

### 4.3 Reproduction of Paper Figure 1

Figure 3 reproduces the three-panel diagnostic from Figure 1 of [2], using the kernel-weighted complex order parameter from Equation 3.

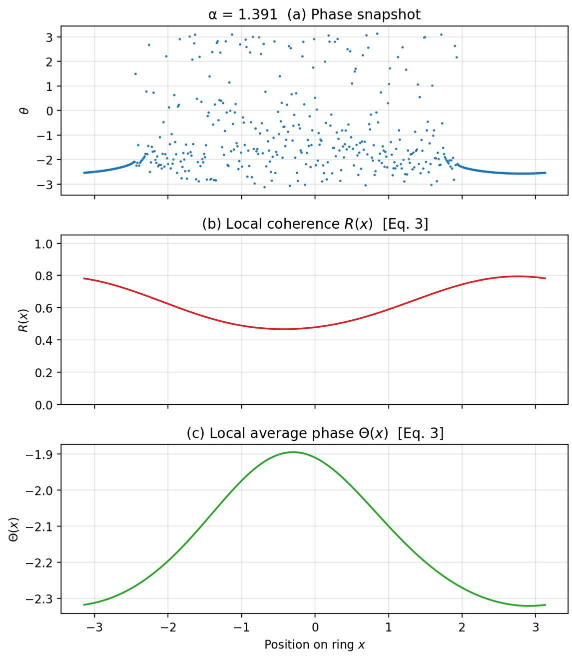


*Figure 3: Reproduction of Figure 1 from [2]. (a) Phase snapshot $\theta(x)$. (b) Local coherence $R(x)$. (c) Local average phase $\Theta(x)$.*

**Panel (a)** shows the phase snapshot at $t = 5000$. Locked oscillators near $x \approx \pm\pi$ lie on a smooth arc at $\theta \approx -2.5$; drifting oscillators in the middle are scattered across all phases and cluster more densely near the arc because they slow down as they pass through the coherent pack [2].

**Panel (b)** shows the kernel-weighted $R(x)$ from Equation 3. Unlike the sliding-window $|z(x)|$ in Figure 2, this uses the coupling kernel $G(x)$ as the weighting function, matching the paper's exact definition. The profile is smooth and cosine-like, ranging from $R \approx 0.78$ at the ring boundaries to $R \approx 0.48$ in the incoherent center. This shape is predicted by the exact solution Equation 4.

**Panel (c)** shows the local average phase $\Theta(x)$, varying smoothly from $\Theta \approx -2.3$ at the edges to $\Theta \approx -1.9$ at the center, consistent with the cosine profile predicted by Equation 5.

### *4.4 Mean Drift Frequency*

Figure 4 shows the time-averaged angular velocity $\langle\dot{\theta}\rangle$ of each oscillator, computed from unwrapped phases over the second half of the simulation.

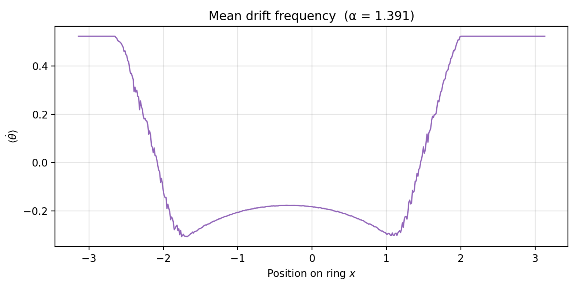


*Figure 4: Mean drift frequency $\langle\dot{\theta}\rangle(x)$ for each oscillator.*

The oscillators split into two distinct groups:

- **Locked oscillators** (near $x \approx \pm\pi$): All share the same time-averaged frequency $\langle\dot{\theta}\rangle \approx 0.52$, forming a flat plateau. This common frequency is the collective rotation rate of the coherent domain.
- **Drifting oscillators** (middle): Their time-averaged frequency drops smoothly to $\langle\dot{\theta}\rangle \approx -0.2$ near $x \approx 0$, forming a bowl-shaped profile. The smooth variation within the drifting region reflects how much time each drifter spends near the locked pack depending on its position.

The transition between the locked plateau and the drifting bowl occurs at $x \approx \pm 2$, which matches the boundary where $R(x)$ falls below the critical threshold in the paper's self-consistency analysis [2]. Identical oscillators under the same coupling end up with measurably different long-term frequencies depending solely on their position on the ring.

### *4.5 Bifurcation Analysis*

Figure 5 shows the time-averaged global order parameter $\langle R\rangle$ as a function of $\alpha$, from a sweep of 20 evenly spaced values in $[1.2, \pi/2]$.

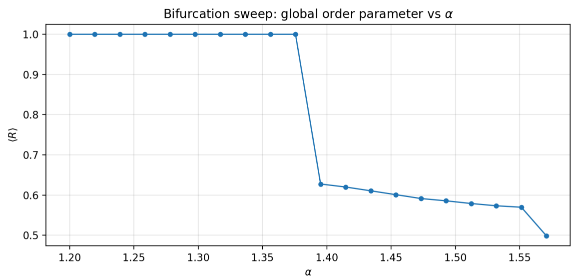


*Figure 5: Bifurcation sweep: global order parameter $\langle R\rangle$ vs. phase-lag $\alpha$.*

The sweep reveals three distinct regimes:

- **Full synchronization** ($\alpha \lesssim 1.37$): $\langle R\rangle \approx 1.0$, meaning all oscillators are phase-locked and the synchronized state is the only stable attractor.
- **Critical transition** ($\alpha \approx 1.38$): A sharp drop from $\approx 1.0$ to $\approx 0.63$ over a very narrow range of $\alpha$. This is where the chimera state is born.
- **Chimera regime** ($\alpha \gtrsim 1.39$): The order parameter sits around $\langle R\rangle \approx 0.57$–$0.63$, reflecting the stable mixture of locked and drifting oscillators. The intermediate value (neither 0 nor 1) is the chimera's fingerprint.
- **Approach to full drift** ($\alpha \to \pi/2$): $\langle R\rangle$ gradually drops further toward $\approx 0.50$ as the coupling becomes increasingly anti-phase.

The critical value $\alpha_c \approx 1.38$ corresponds to $\beta_c = \pi/2 - \alpha_c \approx 0.19$, consistent with the chimera existence boundary predicted by [2] for $A = 0.995$. In the paper, the chimera disappears via a saddle-node bifurcation at a maximum $\beta_{\max}$ that depends on $A$. Our numerical sweep captures this transition cleanly.

## 5. Discussion

### *5.1 Comparison with the Original Paper*

Our results are in good qualitative agreement with the analytical predictions and simulations of [2]:

- The phase snapshot (Figure 3, panel a) matches the paper's Figure 1a: locked oscillators near $x = \pm\pi$, drifting oscillators in the middle.
- The kernel-weighted $R(x)$ profile (panel b) is smooth and cosine-like, as predicted by Equation 4, with minimum $R \approx 0.48$.
- The $\Theta(x)$ profile (panel c) shows the expected cosine variation from Equation 5.
- The drift frequency (Figure 4) shows the characteristic flat-plateau-plus-bowl structure with a sharp boundary between locked and drifting populations.
- The bifurcation sweep (Figure 5) places the sync-to-chimera transition at the expected $\alpha$ value.

### *5.2 Finite-N Effects*

Several features of our results come from the finite oscillator count:

- **Jagged $|z(x)|$**: The sliding-window order parameter (Figure 2) fluctuates in the incoherent region because the local average is over a finite window of 81 oscillators. The kernel-weighted $R(x)$ (Figure 3, panel b) is smoother because it uses the full coupling kernel as the weight function.

- **Chimera metastability**: In the $N \to \infty$ limit, chimera states are true attractors. For finite $N$ they are metastable: the chimera will eventually collapse, though the collapse time grows exponentially with $N$. The chimera-survival check (Section 3.3) deals with this by catching early collapses and retrying.

- **Sensitivity to initial conditions**: Not every random perturbation nucleates a chimera. The retry mechanism typically needs 1–3 attempts, which confirms that chimera nucleation is genuinely sensitive to the initial kick.

### *5.3 Numerical Considerations*

The original paper uses a fixed-step RK4 integrator at $\delta t = 0.025$ and initial condition width $\kappa = 30$. Our adaptive RK45 solver required two adjustments:

1. **Wider initial perturbation** ($\kappa = 0.76$ vs. 30): The narrow Gaussian spike with $\kappa = 30$ was routinely absorbed into the synchronized state before the chimera could form. The adaptive solver's variable step size smooths out very localized features in the early transient, so a wider initial kick is needed to seed incoherence over a large enough spatial extent.

2. **Capped maximum step** (max_step = 0.1): Without this, the solver occasionally took steps large enough to skip over the fast phase dynamics during chimera formation. The cap of 0.1 keeps temporal resolution adequate while still benefiting from adaptive error control.

## 6. Conclusion

In summary, we successfully simulated chimera states in the Abrams–Strogatz model, showing that a ring of 512 identical oscillators can spontaneously split into a coherent (phase-locked) domain and an incoherent (drifting) domain with no heterogeneity required. Our quantitative diagnostics (local coherence $R(x)$, local average phase $\Theta(x)$, and mean drift frequency $\langle\dot{\theta}\rangle(x)$) closely match the exact analytical predictions of [2], and the reproduced three-panel figure agrees well with the original paper's Figure 1.

Additionally, our bifurcation sweep over $\alpha$ located the critical transition at $\alpha_c \approx 1.38$, where $\langle R\rangle$ drops sharply from $\approx 1$ to $\approx 0.6$, consistent with the saddle-node bifurcation predicted by [2]. Dealing with finite-size effects also showed that while the chimera is stable once formed, its nucleation is highly sensitive to initial conditions, requiring an automated retry mechanism and careful tuning of the solver. Ultimately, replicating this model demonstrates how nonlocal coupling provides the spatial degrees of freedom necessary for coherent and incoherent domains to coexist. Future work could explore how these states behave under different coupling topologies or in two-dimensional arrays.

## 6. Bibliography


[1] Y. Kuramoto and D. Battogtokh, "Coexistence of Coherence and Incoherence in Nonlocally Coupled Phase Oscillators," *Nonlinear Phenomena in Complex Systems*, vol. 5, no. 4, pp. 380–385, 2002.

[2] D. M. Abrams and S. H. Strogatz, "Chimera States for Coupled Oscillators," *Physical Review Letters*, vol. 93, no. 17, p. 174102, 2004.

[3] A. T. Winfree, "Biological Rhythms and the Behavior of Populations of Coupled Oscillators," *Journal of Theoretical Biology*, vol. 16, no. 1, pp. 15–42, 1967.

[4] Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence*. Berlin: Springer-Verlag, 1984.

[5] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed. Boulder, CO: Westview Press, 2015.

[6] S.-i. Shima and Y. Kuramoto, "Rotating Spiral Waves with Phase-Randomized Core in Nonlocally Coupled Oscillators," *Physical Review E*, vol. 69, no. 3, p. 36213, 2004.

[7] M. R. Tinsley, S. Nkomo, and K. Showalter, "Chimera and Phase-Cluster States in Populations of Coupled Chemical Oscillators," *Nature Physics*, vol. 8, no. 9, pp. 662–665, 2012.

[8] N. C. Rattenborg, C. J. Amlaner, and S. L. Lima, "Behavioral, Neurophysiological and Evolutionary Perspectives on Unihemispheric Sleep," *Neuroscience & Biobehavioral Reviews*, vol. 24, no. 8, pp. 817–842, 2000.